

Q1:

Algorithm 1: Study(t)Input: term t Output: Function returns "true" if Goliath learned term t ; returns "false" otherwise.

begin:

if has_learned = true then

return true

 $E = \text{find_explain}(t)$ if $|E| = 0$ then

return false

if $|E| = 1$ thenmemorize(t)

return true

for each term s in E thenif study(s) is false then

return false

memorize(t)

return true

end

Q2:

mark the time cost for each line:

Algorithm 1: search(A, x)Input: An array A of n integers and one integer x Output: Yes if x is in A , No otherwise.

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1. begin -----  $O(1)$ 
2.   if  $n = 0$  then -----  $O(1)$ 
3.     return No -----  $O(1)$ 
4.   else if  $n = 1$  then -----  $O(1)$ 
5.     if  $A[0] = x$  then ----  $O(1)$ 
6.       return Yes ----  $O(1)$ 
7.     else -----  $O(1)$ 
8.       return No ---  $O(1)$ 
9.     end if ----  $O(1)$ 
10.  else -----  $O(1)$ 
11.    left =  $A[0 \dots \frac{n}{2}]$  -----  $T(\frac{n}{2})$ 
12.    right =  $A[\frac{n}{2} + 1 \dots n - 1]$  --  $T(\frac{n}{2})$ 
13.    return search(left,  $x$ ) OR search(right,  $x$ ) --  $O(1)$ 
14.  end if -----  $O(1)$ 
15. end
```

Time complexity:

$$T(n) = \begin{cases} O(1) & , n = 1 \text{ or } 0 \\ 2T(\frac{n}{2}) + O(1), & n > 1 \end{cases}$$

when $n > 1$,

$$T(n) = 2T(\frac{n}{2}) + cn = 2[2T(\frac{n}{4}) + \frac{cn}{2}] + cn$$

$$= \dots = 2^k T(\frac{n}{2^k}) + kcn$$

$$\frac{n}{2^k} = 1, \quad k = \log_2 n$$

$$T(n) = 2^{\log_2 n} T(1) + (\log_2 n)cn$$

$$= cn + cn \log_2 n = O(n \log n)$$

$$\text{thus } T(n) = \begin{cases} O(1), & n = 1 \text{ or } 0 \\ O(n \log n), & n > 1 \end{cases}$$

Q3:

(a) $f(1)=1$, $f(x)=f(x-1)+x$ for $x \geq 2$.

$$f(x) = f(x-1) + x$$

$$= [f(x-2) + (x-1)] + x$$

$$= [f(x-3) + (x-2)] + (x-1) + x$$

$$= f(1) + 2 + 3 + \dots + x-1 + x$$

$$= 1 + 2 + 3 + \dots + x-1 + x$$

$$= \frac{(1+x)x}{2}$$

$$\text{check: } f(x-1) + x$$

$$= \frac{(1+x-1)(x-1)}{2} + x$$

$$= \frac{x(x-1) + 2x}{2}$$

$$= \frac{x+x^2}{2}$$

$$= \frac{(1+x)x}{2}$$

$$= f(x)$$

when $x=1$

$$f(1) = \frac{(1+1) \times 1}{2} = 1$$

So the result is $f(x) = \frac{x(1+x)}{2}$

(b) $f(1)=2$, $f(x)=2f(\frac{x}{2})+x+1$ for $x \geq 2$

$$f(x) = 2f(\frac{x}{2}) + x + 1$$

$$= 2[2f(\frac{x}{2^2}) + \frac{x}{2} + 1] + x + 1 = 2^2 f(\frac{x}{2^2}) + 2x + (1+2)$$

$$= 2^3 [2f(\frac{x}{2^3}) + \frac{x}{2^2} + 1] + 2x + 3 = 2^3 f(\frac{x}{2^3}) + 3x + (1+2+2^2)$$

$$= 2^4 [2f(\frac{x}{2^4}) + \frac{x}{2^3} + 1] + 3x + 4 = 2^4 f(\frac{x}{2^4}) + 4x + (1+2+2^2+2^3)$$

$$= 2^n f(\frac{x}{2^n}) + nx + \frac{1 - (\frac{1}{2})^n}{1 - \frac{1}{2}}$$

$$= 2^n f(\frac{x}{2^n}) + nx + 2 - 2 \cdot (\frac{1}{2^n})$$

let $x = 2^n$, then $n = \log_2 x$

$$\text{then } f(x) = x f(1) + x \log_2 x + 2 - 2 \cdot \frac{1}{x}$$

$$= x \log_2 x + 2x - 2 \cdot \frac{1}{x} + 2$$

$$\text{check: when } x=1: f(1) = 0 + 2 - 2 + 2 = 2$$

so the result is $f(x) = x \log_2 x + 2x - \frac{2}{x} + 2$

Q4: let $f(x) = \frac{x^2}{6} + 99x$, prove: $f(x) = O(x^2)$

Proof:

By definition of "O", need to prove: $\frac{x^2}{6} + 99x \leq Cx^2$ ($x \geq x_0, x > 0$)

for every $x \geq 1$, has $x^2 \geq x$

$$\text{then: } \frac{x^2}{6} + 99x \leq \frac{x^2}{6} + 99x^2 = \frac{595}{6}x^2 \quad (x \geq 1)$$

let $C = \frac{595}{6}$, $x_0 = 1$, then have $x \geq x_0 = 1$:

$$\square \text{ satisfy: } f(x) = \frac{x^2}{6} + 99x \leq \frac{595}{6}x^2 \text{ for all } x \geq x_0 = 1$$

so $f(x) = O(x^2)$

Q5: $f(0)=1, f(1)=3, f(x)=f(x-1)+f(x-2)$ for all $x \geq 2$
prove $f(x) = O(2^x)$

$$\text{when: } x=0, f(0)=1 \leq 2 \cdot 2^0 = 2, \\ x=1, f(1)=3 \leq 2 \cdot 2^1 = 4$$

both satisfy $f(x) = O(2^x)$

assume: when $k \geq 2$ ($k \in \mathbb{N}$):

$$f(k-1) \leq 2 \cdot 2^{k-1} = 2^k, \quad f(k-2) \leq 2 \cdot 2^{k-2} = 2^{k-1}$$

need to prove when $x=k$, $f(k) \leq 2^{k+1}$ is true

$$\begin{aligned} f(x) &= f(k-1) + f(k-2) \\ &\leq 2^k + 2^{k-1} \\ &= 2^{k-1}(2+1) \\ &= 3 \cdot 2^{k-1} = \frac{3}{2} \cdot 2^k \\ &\leq 2^{k+1} \end{aligned}$$

Then, when $x=k$, $f(k) \leq 2^{k+1}$ is true

$\square$ so, using induction for all $x \geq 0$ ($x \in \mathbb{N}$), have $f(x) \leq 2^{x+1} = 2 \cdot 2^x$
($C=2, x_0=0$) satisfy $f(x) = O(2^x)$

Q6 =

(a) since $f(x) = o(x)$, $g(x) = o(x^2)$:

$$|f(x)| \leq C_1 \cdot x, |g(x)| \leq C_2 \cdot x^2$$

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \\ \leq C_1 \cdot x + C_2 \cdot x^2$$

for all $x > 1$ have $x^2 > x$, so $C_1 x^2 > C_1 x$

$$\text{then } |f(x) + g(x)| \leq C_1 x + C_2 x^2 < C_1 x^2 + C_2 x^2 = (C_1 + C_2) x^2$$

so for all $C = C_1 + C_2$, $x_0 = 1$, have $|f(x) + g(x)| \leq C x^2$
thus $f(x) + g(x) = o(x^2)$

(b) $|f(x) \times g(x)| = |f(x)| \times |g(x)|$

$$\leq C_1 \cdot x \times C_2 \cdot x^2$$

$$= (C_1 \times C_2) x^3$$

$$\text{Let } C = C_1 \times C_2,$$

then for all $C = C_1 \times C_2$, $x \geq x_0$ have $|f(x) \times g(x)| \leq C x^3$

thus $f(x) \times g(x) = o(x^3)$

(c) $f(g(x)) = o(x^2)$ means for all $g(x) \geq x_1$ exist $f(g(x)) \leq C x^2$

since $g(x) = o(x^2)$; when $x \geq x_2$, $|g(x)| \leq C_2 \cdot x^2$

so $g(x)$ can larger than x_1 when x is big enough.

$$|f(g(x))| \leq C_1 |g(x)| \leq C_1 \cdot C_2 \cdot x^2$$

then for all $C = C_1 \cdot C_2$, $g(x) \geq x_1$, have $|f(g(x))| \leq C \cdot x^2$

thus $f(g(x)) = o(x^2)$